KAIZEN MATHS

Lesson starter

Correlation and Regression

Regression Line

Name _____

SA1

The table shows delivery distance (km) and delivery time (minutes) for 6 observations. Find the least-squares regression line of y on x.

Observation	x	y	x^2	xy
1	2	17	4	34
2	4	26	16	104
3	6	28	36	168
4	8	37	64	296
5	10	45	100	450
6	12	47	144	564
Total	42	200	364	1616

Show your working

KAIZEN MATHS

Lesson starter

Correlation and Regression

Regression Line

Name _____

SB1

For a data set comparing delivery distance and delivery time, the summary statistics are given below. Find the regression line of y on x.

n	6
sum x	42
sum y	200
sum x^2	364
sum xy	1616
sum y^2	7352

Show your working

KAIZEN MATHS

Lesson starter

Correlation and Regression

Regression Line

Name _____

SC1

For advertising spend and weekly sales, the regression line of y on x is $\hat{y} = 22.000 + 9.000x$. Interpret the gradient and intercept in context.

Show your working

KAIZEN MATHS

Lesson starter

Correlation and Regression

Regression Line

Name _____

SD1

The table shows age of car (years) and resale value (thousands of pounds) for 6 observations. Find the least-squares regression line of y on x.

Observation	x	y	x^2	xy
1	1	23	1	23
2	2	19	4	38
3	3	18	9	54
4	4	14	16	56
5	5	15	25	75
6	6	11	36	66
Total	21	100	91	312

Show your working

KAIZEN MATHS

Exit ticket

Correlation and Regression

Regression Line

Name _____

EA1

For a data set comparing age of car and resale value, the summary statistics are given below. Find the regression line of y on x.

n	6
sum x	21
sum y	100
sum x^2	91
sum xy	312
sum y^2	1756

Show your working

KAIZEN MATHS

Exit ticket

Correlation and Regression

Regression Line

Name _____

EB1

For age of car and resale value, the regression line of y on x is $\hat{y} = 24.267 - 2.171x$. Interpret the gradient and intercept in context.

Show your working

KAIZEN MATHS

Exit ticket

Correlation and Regression

Regression Line

Name _____

EC1

The table shows advertising spend (hundreds of pounds) and weekly sales (items) for 6 observations. Find the least-squares regression line of y on x.

Observation	x	y	x^2	xy
1	2	41	4	82
2	3	46	9	138
3	4	60	16	240
4	5	66	25	330
5	6	79	36	474
6	7	83	49	581
Total	27	375	139	1845

Show your working

KAIZEN MATHS

Exit ticket

Correlation and Regression

Regression Line

Name _____

EA1

For a data set comparing age of car and resale value, the summary statistics are given below. Find the regression line of y on x.

n	6
sum x	21
sum y	100
sum x^2	91
sum xy	312
sum y^2	1756

Show your working

Teacher answers & working

Correlation and Regression · Regression Line

STA-00F2-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1

The table shows delivery distance (km) and delivery time (minutes) for 6 observations. Find the least-squares regression line of y on x.

Observation	x	y	x^2	xy
1	2	17	4	34
2	4	26	16	104
3	6	28	36	168
4	8	37	64	296
5	10	45	100	450
6	12	47	144	564
Total	42	200	364	1616

Answer:

$\hat{y} = 11.733 + 3.086x$

Working:

Find the summary values from the table.

$\Sigma x = 42, \Sigma y = 200, \Sigma x^2 = 364, \Sigma xy = 1616, n = 6$

Calculate $\bar{x}$, $\bar{y}$, Sxx and Sxy.

$\bar{x} = \frac{42}{6} = 7.000$

$\bar{y} = \frac{200}{6} = 33.333$

$Sxx = \Sigma x^2 - \frac{(\Sigma x)^2}{n} = 364 - \frac{42^2}{6} = 70.000$

$Sxy = \Sigma xy - \frac{(\Sigma x)(\Sigma y)}{n} = 1616 - \frac{42 \times 200}{6} = 216.000$

Find the gradient and intercept of the regression line of y on x.

$b = \frac{Sxy}{Sxx} = \frac{216.000}{70.000} = 3.086$

$a = \bar{y} - (b \times \bar{x}) = 33.333 - (3.086 \times 7.000) = 11.733$

$\hat{y} = 11.733 + 3.086x$

Teacher answers & working

Correlation and Regression · Regression Line

STA-00F2-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1

For a data set comparing delivery distance and delivery time, the summary statistics are given below. Find the regression line of y on x.

n	6
sum x	42
sum y	200
sum x^2	364
sum xy	1616
sum y^2	7352

Answer:

$\hat{y} = 11.733 + 3.086x$

Working:

Find the summary values from the table.

$\Sigma x = 42$, $\Sigma y = 200$, $\Sigma x^2 = 364$, $\Sigma xy = 1616$, $n = 6$

Calculate $\bar{x}$, $\bar{y}$, Sxx and Sxy.

$\bar{x} = \frac{42}{6} = 7.000$

$\bar{y} = \frac{200}{6} = 33.333$

$Sxx = \Sigma x^2 - \frac{(\Sigma x)^2}{n} = 364 - \frac{42^2}{6} = 70.000$

$Sxy = \Sigma xy - \frac{(\Sigma x)(\Sigma y)}{n} = 1616 - \frac{42 \times 200}{6} = 216.000$

Find the gradient and intercept of the regression line of y on x.

$b = \frac{Sxy}{Sxx} = \frac{216.000}{70.000} = 3.086$

$a = \bar{y} - (b \times \bar{x}) = 33.333 - (3.086 \times 7.000) = 11.733$

$\hat{y} = 11.733 + 3.086x$

Teacher answers & working

Correlation and Regression · Regression Line

STA-00F2-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1

For advertising spend and weekly sales, the regression line of y on x is ŷ = 22.000 + 9.000x. Interpret the gradient and intercept in context.

Answer:

Gradient: 9.000

Working:

Identify the gradient and intercept.

$a = 22.000$ and $b = 9.000$ in $\hat{y} = a + bx$.

Interpret the gradient.

For each increase of 1 hundreds of pounds in advertising spend, the predicted weekly sales changes by about 9.000 items.

Interpret the intercept cautiously.

The intercept is the predicted weekly sales when advertising spend is 0, but it may not be meaningful if 0 is outside the observed range.

Teacher answers & working

Correlation and Regression · Regression Line

STA-00F2-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1

The table shows age of car (years) and resale value (thousands of pounds) for 6 observations. Find the least-squares regression line of y on x.

Observation	x	y	x^2	xy
1	1	23	1	23
2	2	19	4	38
3	3	18	9	54
4	4	14	16	56
5	5	15	25	75
6	6	11	36	66
Total	21	100	91	312

Answer:

$\hat{y} = 24.267 - 2.171x$

Working:

Find the summary values from the table.

$\Sigma x = 21, \Sigma y = 100, \Sigma x^2 = 91, \Sigma xy = 312, n = 6$

Calculate $\bar{x}$, $\bar{y}$, Sxx and Sxy.

$\bar{x} = \frac{21}{6} = 3.500$

$\bar{y} = \frac{100}{6} = 16.667$

$Sxx = \Sigma x^2 - \frac{(\Sigma x)^2}{n} = 91 - \frac{21^2}{6} = 17.500$

$Sxy = \Sigma xy - \frac{(\Sigma x)(\Sigma y)}{n} = 312 - \frac{21 \times 100}{6} = -38.000$

Find the gradient and intercept of the regression line of y on x.

$b = \frac{Sxy}{Sxx} = \frac{-38.000}{17.500} = -2.171$

$a = \bar{y} - (b \times \bar{x}) = 16.667 - (-2.171 \times 3.500) = 24.267$

$\hat{y} = 24.267 - 2.171x$

Teacher answers & working

Correlation and Regression · Regression Line

STA-00F2-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1

For a data set comparing age of car and resale value, the summary statistics are given below. Find the regression line of y on x.

n	6
sum x	21
sum y	100
sum x^2	91
sum xy	312
sum y^2	1756

Answer:

$\hat{y} = 24.267 - 2.171x$

Working:

Find the summary values from the table.

$\Sigma x = 21, \Sigma y = 100, \Sigma x^2 = 91, \Sigma xy = 312, n = 6$

Calculate $\bar{x}$, $\bar{y}$, Sxx and Sxy.

$\bar{x} = \frac{21}{6} = 3.500$

$\bar{y} = \frac{100}{6} = 16.667$

$Sxx = \Sigma x^2 - \frac{(\Sigma x)^2}{n} = 91 - \frac{21^2}{6} = 17.500$

$Sxy = \Sigma xy - \frac{(\Sigma x)(\Sigma y)}{n} = 312 - \frac{21 \times 100}{6} = -38.000$

Find the gradient and intercept of the regression line of y on x.

$b = \frac{Sxy}{Sxx} = \frac{-38.000}{17.500} = -2.171$

$a = \bar{y} - (b \times \bar{x}) = 16.667 - (-2.171 \times 3.500) = 24.267$

$\hat{y} = 24.267 - 2.171x$

EB1

For age of car and resale value, the regression line of y on x is $\hat{y} = 24.267 - 2.171x$. Interpret the gradient and intercept in context.

Answer:

Gradient: -2.171

Working:

Identify the gradient and intercept.

$a = 24.267$ and $b = -2.171$ in $\hat{y} = a + bx$.

Interpret the gradient.

For each increase of 1 years in age of car, the predicted resale value changes by about -2.171 thousands of pounds.

Interpret the intercept cautiously.

The intercept is the predicted resale value when age of car is 0, but it may not be meaningful if 0 is outside the observed range.

Teacher answers & working

Correlation and Regression · Regression Line

STA-00F2-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1

The table shows advertising spend (hundreds of pounds) and weekly sales (items) for 6 observations. Find the least-squares regression line of y on x.

Observation	x	y	x^2	xy
1	2	41	4	82
2	3	46	9	138
3	4	60	16	240
4	5	66	25	330
5	6	79	36	474
6	7	83	49	581
Total	27	375	139	1845

Answer:

$\hat{y} = 22.000 + 9.000x$

Working:

Find the summary values from the table.

$\Sigma x = 27, \Sigma y = 375, \Sigma x^2 = 139, \Sigma xy = 1845, n = 6$

Calculate $\bar{x}$, $\bar{y}$, Sxx and Sxy.

$\bar{x} = \frac{27}{6} = 4.500$

$\bar{y} = \frac{375}{6} = 62.500$

$Sxx = \Sigma x^2 - \frac{(\Sigma x)^2}{n} = 139 - \frac{27^2}{6} = 17.500$

$Sxy = \Sigma xy - \frac{(\Sigma x)(\Sigma y)}{n} = 1845 - \frac{27 \times 375}{6} = 157.500$

Find the gradient and intercept of the regression line of y on x.

$b = \frac{Sxy}{Sxx} = \frac{157.500}{17.500} = 9.000$

$a = \bar{y} - (b \times \bar{x}) = 62.500 - (9.000 \times 4.500) = 22.000$

$\hat{y} = 22.000 + 9.000x$